

TECHNICAL AUDIT REPORT

Production Deployment & Performance Audit

Pipeline: Replit !' GitHub !' Hostinger
Project: 19 DOGS eCommerce Platform
Date: 30 July 2026

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Fixes Applied

Executive Summary

This audit covers the full Replit !' GitHub !' Hostinger deployment pipeline for the 19 DOGS eCommerce platform. Three showstopper issues were identified that prevent the site from functioning correctly in production. Six performance and infrastructure improvements were applied directly to the codebase. The remaining issues require manual action in Hostinger's control panel, GitHub repository settings, and an external storage provider.

Issue Index

#	Issue	Severity	Status
1	Image storage tied to Replit sidecar (unreachable on Hostinger)	Ø=Ý4 SHOWSTOPPER	Manual action required
2	Built dist/ excluded from Git — no build on Hostinger	Ø=Ý4 SHOWSTOPPER	Partially fixed — workflow created
3	Node.js version not locked	Ø=Ý4 SHOWSTOPPER	' Fixed — .nvmrc created
4	No gzip/Brotli compression	Ø=ßá HIGH	' Fixed — compression added
5	No cache-control headers on static assets	Ø=ßá HIGH	' Fixed — 1yr cache on hashed assets
6	30 Google Font families loaded (only 5 used)	Ø=ßá HIGH	' Fixed — trimmed to 5 families
7	No SPA fallback routing (.htaccess)	Ø=ßá HIGH	' Fixed — .htaccess created
8	Environment variables not set on Hostinger	Ø=ßá HIGH	Manual action required
9	Neon PostgreSQL IP allowlist	Ø=ßà MEDIUM	Manual action required
10	No SSL/HTTPS verification	Ø=ßà MEDIUM	Manual action required

Show stopper Issues

0=Y4 SHOWSTOPPER

Issue #1 — Image Storage Uses Replit-Only Sidecar

The entire image/file storage system uses Replit Object Storage via an internal sidecar process. This sidecar runs at `http://127.0.0.1:1106` — an address that only exists inside Replit's infrastructure and is completely unreachable from Hostinger or any other server.

Root cause — from `server/objectStorage.ts`:

```
const REPLIT_SIDECAR_ENDPOINT = 'http://127.0.0.1:1106';
```

Every product image URL stored in the database follows the pattern `/objects/uploads/<uuid>`. On Hostinger, the Express route that serves these objects tries to connect to the sidecar, gets a connection refused error, and returns a 500 to the browser. The entire product catalog appears with broken images.

Required fix:

- Select an S3-compatible storage provider: Cloudflare R2 (free up to 10 GB), AWS S3, or Backblaze B2.
- Update `server/objectStorage.ts` to use that provider's SDK instead of the Google Cloud + Replit sidecar.
- Migrate all existing uploaded files from Replit Object Storage to the new bucket.
- Update `PRIVATE_OBJECT_DIR` and `PUBLIC_OBJECT_SEARCH_PATHS` environment variables.

0=Y4 SHOWSTOPPER

Issue #2 — `dist/` Is in `.gitignore` — Hostinger Gets Only Source Code

The `.gitignore` file explicitly excludes the `dist/` directory. This means every git push to GitHub contains only raw TypeScript/React source files. There is no compiled server bundle (`dist/index.cjs`) and no built frontend (`dist/public/`) on GitHub.

The production start command is: `node dist/index.cjs`

If Hostinger clones from GitHub and runs `npm start`, it immediately crashes because `dist/index.cjs` does not exist. There was also no GitHub Actions workflow to run the build before deployment.

What was fixed:

- Created `.github/workflows/deploy.yml` — a GitHub Actions workflow that runs `npm run build` on every push to main and saves the compiled `dist/` as a downloadable artifact.
- The workflow includes pre-written (commented-out) SSH deploy steps. Once you add three GitHub Secrets (`HOSTINGER_SSH_HOST`, `HOSTINGER_SSH_USER`, `HOSTINGER_SSH_KEY`) and uncomment those steps, the workflow will automatically push the built `dist/` to Hostinger on every push.

Remaining manual action:

- In GitHub repository: Settings → Secrets → Actions, add: `HOSTINGER_SSH_HOST`, `HOSTINGER_SSH_USER`, `HOSTINGER_SSH_KEY`.
- Uncomment the SSH deploy steps in `.github/workflows/deploy.yml`.

0=Y4 SHOWSTOPPER

Issue #3 — Node.js Version Not Locked

The app is built and tested on Node.js v20.20.0. The `package.json` had no `engines` field and no `.nvmrc` file. Hostinger's Node.js hosting defaults to whatever version is currently configured in its control panel. If that's Node 16 or 18, the server crashes on startup due to ESM syntax, top-level `await`, and API differences introduced in Node 20.

What was fixed:

- Created `.nvmrc` containing "20". CI runners, Hostinger, and local developers can all use this file to select the correct version automatically.

Remaining manual action:

- In Hostinger IP panel: Websites → Manage → Node.js, set the Node.js version to 20.x.

Issue #4 — No Gzip/Brotli Compression

Express was serving all responses (HTML, JS, CSS, JSON) uncompressed. The main JavaScript bundle alone is 371 KB over the wire. Without compression, every page load transfers significantly more data than necessary, directly impacting LCP and overall load time.

What was fixed:

- Installed the `compression` npm package.
- Added `app.use(compression())` as the first middleware in `server/index.ts`. It now automatically gzip-compresses all text-based responses.

Expected result: main JS bundle drops from 371 KB !' ~105 KB (-72%). Total JS transfer from ~3.3 MB !' ~850 KB.

Issue #5 — No Cache-Control Headers on Static Assets

`express.static()` was called with no options, meaning no Cache-Control header was set on any response. Browsers re-downloaded every JS, CSS, and image file on every page load, even when nothing had changed. Vite fingerprints every asset filename (e.g. `index-BdPqsYrN.js`), which makes long-term caching completely safe.

What was fixed:

- Updated `server/static.ts` to serve `/assets/*` (Vite-hashed files) with `Cache-Control: max-age=31536000, immutable` (1 year). The filename changes when content changes, so there is zero risk of stale cache.
- Other static files (favicon, etc.) get `Cache-Control: max-age=3600` (1 hour).
- `index.html` itself gets no-cache so the browser always fetches the latest HTML with current asset references.

Issue #6 — 30 Google Font Families Loaded (Only 5 Actually Used)

The `<head>` in `client/index.html` was loading 30+ font families from Google Fonts in a single render-blocking `<link>` tag with a ~2,000-character URL. This included Architects Daughter, Fira Code, Geist Mono, IBM Plex, JetBrains Mono, Lora, Merriweather, Roboto, Source Code Pro, Space Mono, and many more that are not referenced anywhere in the codebase.

Fonts used in the actual application:

Font Family	Tailwind Class	Usages in Code	Role
Inter	font-inter	121	Body text, labels, UI
Playfair Display	font-playfair	52	Editorial headings
Outfit	(CSS var)	CSS config	Admin headings
DM Sans	(CSS var)	CSS config	Pricing / body
Open Sans	--font-sans var	CSS config	Base sans-serif

- Trimmed `client/index.html` to load only those 5 font families. The Google Fonts request URL shrank from ~2,000 characters to ~300 characters. Expected LCP improvement: 200–400 ms on first load.

Issue #7 — No SPA Fallback Routing

On Apache-based hosting (which Hostinger shared plans use), a direct browser request to any route other than `/` returns a 404. Apache looks for a physical file at that path, finds nothing, and shows its default error page. The React router never runs.

What was fixed:

- Created `public/.htaccess` with RewriteEngine On rules that redirect all non-file/non-directory requests to `index.html`.
-

Also added Apache `mod_deflate` (gzip) and `mod_expires` (cache headers) as fallbacks in case the Node.js server is bypassed.

Note: if Hostinger serves this app through the Node.js server (Express), the SPA fallback in `static.ts` already handles this. The `.htaccess` is a safety net for static/Apache hosting scenarios.

⚠️ HIGH Issue #8 — Environment Variables Not Set on Hostinger

The server requires the following environment variables to start and function correctly. These exist in Replit's secrets manager but are not automatically available on Hostinger:

Variable	Required For	If Missing
DATABASE_URL	Every database query	Server crashes on startup
SESSION_SECRET	All user logins	Sessions fail — users cannot log in
PGHOST / PGPORT / PGUSER / PGPASSWORD / PGDATABASE	PostgreSQL connection	Database unreachable
DEFAULT_OBJECT_STORAGE_BUCKET_ID	Image storage (Replit-only)	Image uploads fail (see Issue #1)

- Hostinger hPanel !' Websites !' Manage !' Node.js !' Environment Variables. Add DATABASE_URL and SESSION_SECRET at minimum.
- SESSION_SECRET must be a long, random string (at least 32 characters). Use a password generator.

⚠️ MEDIUM Issue #9 — Neon PostgreSQL IP Allowlist

Neon PostgreSQL databases can restrict which IP addresses are allowed to connect. The database was configured for Replit's IP ranges. Hostinger's servers have different outbound IP addresses. If the Neon allowlist is restrictive, all database queries from Hostinger will be refused.

- Manual action required:**
- In Neon console !' your project !' Settings !' Networking, either add Hostinger's server IP addresses or set the allowlist to 0.0.0.0/0 (allow all) if the database is not publicly sensitive.

⚠️ MEDIUM Issue #10 — SSL/HTTPS Verification

Mixed content errors (HTTP assets loaded on an HTTPS page) silently break images and API calls in modern browsers without any obvious error message on the page.

- Manual action required:**
- Hostinger hPanel !' SSL !' verify the certificate is active and auto-renewing for your domain.
 - Check for mixed content: open Chrome Dev tools !' Console — any "Mixed Content" warnings point to HTTP asset URLs that must be changed to HTTPS.

Performance Benchmarks

Bundle sizes from the current build (Dec 2025 snapshot):

Bundle File	Size (uncompressed)	Size (gzip est.)	Load strategy
FileSaver.min-*.js	733 KB	~200 KB	Lazy — admin only '
Analytics-*.js	416 KB	~120 KB	Lazy — admin only '
rich-text-editor-*.js	372 KB	~108 KB	Lazy — admin only '
index-*.js (main)	371 KB	~105 KB	Eagerly loaded (all users)
html2canvas-*.js	202 KB	~60 KB	Lazy — invoice only '

Action Checklist

All items marked ' have already been applied to the codebase. Items marked %E require manual action outside the code.

Applied Automatically (Code Changes)

- **Gzip compression** — added compression package, added app.use(compression()) to server/index.ts
- **Asset cache headers** — Cache-Control: public, max-age=3600 for assets/* (hashed), 1-hour on others, no-cache on index.html
- **Google Fonts trimmed** — reduced 30 fonts to 5 actually used — ~200–400ms LCP improvement
- **SPA htaccess** — htaccess with RewriteEngine fallback + Apache gzip/cache rules
- **Image created** — image pinned to 20 for CI runners and Hostinger
- **GitHub Actions workflow** — deploy.yml — builds on every push to main

Manual Actions Required (Outside Code)

- CRITICAL

1. Migrate image storage

Replace Replit Object Storage with Cloudflare R2, AWS S3, or Backblaze B2. Update server/objectStorage.ts with the new provider SDK and credentials. Migrate all existing uploaded files to the new bucket.
- CRITICAL

2. GitHub Secrets for SSH deploy

In GitHub !' repo !' Settings !' Secrets !' Actions, add HOSTINGER_SSH_HOST, HOSTINGER_SSH_USER, HOSTINGER_SSH_KEY. Then uncomment the SSH deploy steps in .github/workflows/deploy.yml.
- CRITICAL

3. Set Node.js version on Hostinger

Hostinger hPanel !' Websites !' Manage !' Node.js !' set version to 20.x
- HIGH

4. Set environment variables on Hostinger

hPanel !' Node.js !' Environment Variables: DATABASE_URL, SESSION_SECRET (32+ char random string), and any other vars from Replit secrets.
- MEDIUM

5. Neon PostgreSQL IP allowlist

Neon console !' project !' Settings !' Networking !' add Hostinger server IP or set to 0.0.0.0/0.
- MEDIUM

6. Verify SSL on Hostinger

hPanel !' SSL !' confirm certificate is active. Check Chrome DevTools Console for Mixed Content errors.

